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BEFORE THE PATENT TRIAL AND APPEAL BOARD

Application Number: 13/582,133
Filing Date: August 31, 2012
Appellant(s): CLEMENTS ET AL.

KILKPATRICK TOWNSEND & STOCKTON LLP
1100 PEACHTREE STREET, SUITE 2800
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For Appellant

EXAMINER'S ANSWER

The present application is being examined under the pre-AIA first to invent provisions.

This is in response to the appeal brief filed January 20, 2015 appealing from the Notice of Panel Decision from Pre-Appeal Brief Review mailed October 29, 2014 and the Final Office action mailed April 22, 2014. Appellant's statement as to the status of claims 1-4 and 6-8 as pending and currently rejected, of claim 5 as cancelled, and of claims 9-14 as withdrawn, is accurate. Currently, claims 1-4 and 6-8 are pending, and have been appealed.

(1) Grounds of Rejection to be Reviewed on Appeal

The Appellant's statement of the issues in the brief is correct.[1] **Every ground of rejection** set forth in the Office action dated April 22, 2014 from which the appeal is taken **is being maintained.**

No rejection has been withdrawn. No new ground of rejection has been made.

(2) Grounds of Rejection

The following ground(s) of rejection, reproduced for the record from the Final Office action mailed on April 22, 2014, are applicable to the appealed claims:

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[1] The Examiner notes for the record a typographical error by Appellant regarding Piggott, for which Appellant has written the publication year as 2997 in lieu of 1997.

Claim Rejections - 35 USC § 103

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

> (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negatived by the manner in which the invention was made.

Claims 1-4 and 6-8 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over US 6,406,706 to Haque *et al.* ("Haque", of record), further in view of Piggott *et al.*, Western Austrian Sandalwood Oil" Extraction by Different Techniques and Variations of the Major Components in Different Sections of a Single Tree, Flavour and Fragrance Journal, Vol. 12, 43-56 (1997) ("Piggott"), and Banerjee *et al.*, Modulatory influence of sandalwood oil on mouse hepatic glutathione S-transferase activity and acid soluble sulphydryl level, Cancer Letters, 68 (1993) 105-109 ("Banerjee", of record)-- alone, or in combination with either or all of-- Ikseer Azam, Vol. IV 19th century, Matba Nizami, Kanpur, 1872 AD, p. 309, TKDL identifier BA4/1854C ("Azam", of record), Va gasena, 12th Century- commentator Shaligram Vaisya, Edited Shankar lalji Jain; Khemraj Shrikrishna Das Prakashan, Bombay, Edn. 1996, page 811, TKDL identifier AK11/3505 ("Va gasena", of record) and Thiruvalluvar gnana vettian, 10-15th Century A.D., Ed: Mangadu Vadivel Mudalia, Pub: Parthina Nayakar & sons,

Thirumagal vilakku press, Chennai, Page 272-278, TKDL identifier GP11/20 ("Thiruvalluvar gnana vettian", of record).

Haque discloses the use of α- and β-santalols- found to be the therapeutic ingredients of sandalwood oil- in a composition for the treatment of HPV-induced tumors, such as cancer of the skin and cervix- i.e. for the treatment with an effective amount of both skin and non-skin cancer in a human. (Abstract; col. 1, ll. 22-23; claims 1 and 5). The α- and β-santalols are obtained from at least one Santalum species selected from the group consisting of S. album, S. yasi, S. papuanum and S. spicatum. (claim 3). The α- and β-santalols are formulated for topical application, e.g. as soap with other excipients, i.e. with pharmaceutically acceptable excipients. (col. 3, l. 56- col. 4, l. 4).

Hague discloses treatment with sandalwood oil of skin and cervical cancer (non-skin), but does not explicitly disclose treatment of other specific cancers, per Appellant's claims 6-8.

Banerjee discloses that sandalwood oil from S. album, given by oral administration (5 μl or 15 μl for 10 and 20 days) to mice, shows an enhancement of glutathione S-transferase (GST) activity and of acid-soluble sulfhydryl (SH) levels, which is suggestive of a possible chemopreventive action of sandalwood oil on carcinogenesis. (Abstract). Measurements in the study are done based on extracts from liver tissue. (p. 106). With respect to doses, the study demonstrated that this oral gavage treatment led to significantly higher SH content in animals treated with both dose levels for 10 days, whereas the increase seen with treatment at 15 μl was significant in only the 10-day treatment group. (p. 108, col. 1).

Thus, Banerjee alone discloses an additional specific cancer, per Appellant's claims 6-8. The following prior art references- Azam, Va gasena and Thiruvalluvar gnana vettian- whether applied separately, or in combination with Banerjee, further affirm this conclusion. They all show that sandalwood oil has been used for hundreds of years in India for treatment of cancer broadly, and not just of the particular cancers disclosed in Hague.

Azam discloses a therapeutic compound formulation comprising, *inter alia,* sandalwood from S. album (which encompasses sandalwood oil) together with other pharmaceutically acceptable excipients, which is formulated as oil and is useful in the treatment of cancer. Administration is local as liniment. Azam discloses treatment of cancer broadly. A person of skill in the art would understand this to encompass the particular cancers of Appellant's claims 6-8.

Va gasena discloses a therapeutic compound formulation comprising *inter alia* sandalwood heartwood from S. album (which encompasses sandalwood oil) together with other pharmaceutically acceptable excipients, which is formulated as medicated oil and is useful in tumor treatment. Administration is via the nasal route, in a dose as directed by a physician. Va gasena discloses treatment of cancer broadly. A person of skill in the art would understand this to encompass the particular cancers of Appellant's claims 6-8.

Thiruvalluvar gnana vettian discloses a therapeutic compound formulation comprising *inter alia* sandalwood seed from S. album (which encompasses sandalwood oil) together with other pharmaceutically acceptable excipients, which is formulated as medicated oil and is useful in tumor treatment. The administration is local, as directed by a physician. Thiruvalluvar gnana vettian discloses treatment of cancer broadly. A person of skill in the art would understand this to encompass the particular cancers of Appellant's claims 6-8.

Haque and Banerjee both disclose isolated sandalwood oil, or some of its active ingredients, but do not disclose that such sandalwood oil can be either distilled or extracted, per Appellant's claims as amended.

Piggott discloses that steam distillation, solvent extraction, supercritical fluid extraction and liquid CO2 extraction are all techniques for obtaining sandalwood oil from Western Australian Sandalwood (*S. spicatum*). (Abstract). Piggott further discloses that when the essential oil from *S. spicatum* was first investigated in the 1920s, it was found that it contained up to 35% of α- and β-santalol, i.e. considerable proportions of farnesol were present. A later examination of steam-distilled oil from the trunkwood showed that sesquiterpene alcohols accounted for more than 90% of the content, with the major components being:

> 2(*E*), 6(*E*)-farnesol (31.6%), *epi*-α-bisabolol (10.7%), (*Z*)-α-santalol (9.1%), (*Z*)-nuciferol (6.5%) and (*Z*)-β-santalol (5.4%).

(p. 43, col. 1 and 2). Per Piggott, this suggested that there are different varieties of sandalwood, and that they might vary in different sections of a single tree. (*Id.*).

Therefore, in Piggott the investigators studied extraction by varying: 1) all of the different methods specified above, and 2) selecting different sections of the wood- e.g., branchwood, buttwood. (p. 44, col. 2- p. 45- col. 1). The results showed considerably variance in the extracted ingredients depending on the methods of extractions, and the different sections of wood. (Tables 1, 2 and 3, and discussion pertaining to these tables). *See also* p. 45, col. 2, which provides as follows:

> Also interesting are the percentages of five of the major sesquiterpene components (1–5), *epi*-α-bisabolol (1), (*Z*)-α-santalol (2) 2(*E*),6(*E*)-farnesol (3), (*Z*)-β-santalol (4) and (*Z*)-nuciferol (5), present in each extract. Steam distillation leads to a much greater proportion (54.2%) of these sesquiterpenes than the other extraction techniques (24.4–26.1%), although the relative ratios are similar.

Accordingly, it would have been obvious to a person of skill in the art, based on the combination of Haque, Pigott and Banerjee- alone, or in combination with and either or all of Azam, Va gasena and Thiruvalluvar gnana vettian, to treat different types of cancer with sandalwood oil with a reasonable expectation of success. A person of skill in the art would have been motivated to do so as Haque shows sandalwood oil to treat different types of cancer- e.g. of the skin and of the cervix on topical application, and Banerjee further suggests that oral administration of sandalwood oil has a chemopreventative action on carcinogenesis, as exemplified from results obtained from liver tissue (i.e. non-skin related). A person of skill in the art would have been further motivated to do so as traditional medicine sources from India (Azam, Va gasena and Thiruvalluvar gnana vettian) point to the use of sandalwood for treating cancer broadly, and not just a particular type of cancer. Based on this disclosure, a person of skill in the art would have been motivated to use sandalwood oil for treating a variety of cancers, to include non-skin cancers.

Although both Haque and Banerjee do not explicitly disclose the particular amounts claimed by Appellants, per Appellant's claims as amended, it would have been further obvious to a person of skill in the art to optimize these amounts, in order to arrive at Appellant's claimed range in % (w/w). A person of skill in the art would have been motivated to do so in order to enhance the therapeutic efficacy. Generally, mere optimization of ranges will not support the patentability of subject matter encompassed by the prior art unless there is evidence indicating such concentration or temperature is critical. "When the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimal or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955); "The normal desire of scientists or artisans to improve upon what is already generally known provides the motivation to determine where in a disclosed set of percentage ranges is the optimum combination of percentages." *In re Peterson*, 315 F.3d 1325, 1330 (Fed. Cir. 2003). It has been held that it is within the skills in the art to select optimal parameters, such as amounts of ingredients, in a composition in order to achieve a beneficial effect. *In re Boesch*, 205 USPQ 215 (CCPA 1980). MPEP 2114.04. Moreover, motivation to do so is additionally found in Banerjee, which shows a study with optimizing such ranges, as expressed in μl. A person of skill in the art would have been motivated to estimate such ranges in other units as well (such as claimed by Appellant), guided by the disclosure in

Banerjee that not all doses studied were found to be efficacious for the duration of treatment, with the lower dose showing better efficacy. A person of skill in the art would have been further motivated to do so guided by the disclosure in Piggott, which shows that it is essential for distilled or extracted sandalwood oil to determine such therapeutic ranges, as depending on the technique of extraction used, and the portion of wood used, the amounts of the active ingredients in sandalwood vary significantly. Thus, the skilled artisan would have had at its disclosure at the time of the invention multiple sources independently of one another, and complimentary to one another, establishing motivation to optimize the dose range of sandalwood oil in practicing the claimed method.

Other relevant art

The Examiner also notes for the record the following cumulative prior art- references 2, 6-12, 14, 16-22 of Appellant's IDS dated 1/21/2014.

(2) Response to Argument

A) The Office action properly established a *prima facie* case of <u>obviousness, and Appellant has not met its burden to rebut it</u>

> a) Appellant argues that "[t]he cited references, alone and in combination, fail to disclose or suggest each and every element of the claims" (Ap. Br. at 5-8).

Appellant's arguments have been considered, but not found to be persuasive.

First, Appellant's arguments all fail based on claim construction. As the Federal Circuit's former Chief Judge Rich once remarked: "the name of the game is the claim", Giles S. Rich, *The Extent of the Protection and Interpretation of Claims-American Perspectives*, 21 Int'l Rev. Indus. Prop. & Copyright L. 497, 499, 501 (1990), and thus, let's begin the analysis with the language of independent claim 1 itself. Claim 1 is directed to: "A method of treating a non-skin cancer in a subject, said method comprising administering to the subject **an effective amount of a composition** comprising distilled or extracted sandalwood oil, wherein the subject has a non-skin cancer, and **wherein the concentration of sandalwood oil in the composition is about 0.0002-0.1% sandalwood oil (w/w).**" (emphasis added). Thus, the claim is broadly directed to any effective amount of a composition, wherein this composition has a specified % of sandalwood oil (w/w).

Applicant's specification provides guidance as to what constitutes "an effective amount of a composition" in para [0036] as follows:

The amount of therapeutic agent effective in treating cancer can depend on the nature of the cancer and its associated symptoms, and can be determined by standard clinical techniques. Therefore, these amounts will vary depending on the type of cancer. In addition, in vitro assays can be employed to identify optimal dosage ranges. The precise dose to be employed in the formulation will also depend on the route of administration, and the seriousness of the disease or disorder, and should be decided according to the judgment of the practitioner and each subject's circumstances. For example, the concentration of sandalwood oil in the composition administered to the subject can be from about 0.5 micromolar to about 300 micromolar. In other examples, the concentration of sandalwood oil can be from about 1 micromolar to about 150 micromolar, from about 5 micromolar to about 150 micromolar, from about 10 micromolar to about 150 micromolar, from about 20 micromolar to about 150 micromolar, from about 30 micromolar to about 150 micromolar, from about 40 micromolar to about 150 micromolar, from about 50 micromolar to about 150 micromolar, 60 micromolar to about 150 micromolar, 70 micromolar to about 150 micromolar, 80 micromolar to about 150 micromolar, 90 micromolar to about 150 micromolar, 100 micromolar to 150 micromolar, 125 micromolar to 150 micromolar, 1 micromolar to about 300 micromolar, from about 5 micromolar to about 300 micromolar, from about 10 micromolar to about 300 micromolar, from about 20 micromolar to about 300 micromolar, from about 30 micromolar to about 300 micromolar, from about 40 micromolar to about 300 micromolar, from about 50 micromolar to about 300 micromolar, 60 micromolar to about 300 micromolar, 70 micromolar to about 300 micromolar, 80 micromolar to about 300 micromolar, 90 micromolar to about 300 micromolar, 100 micromolar to 300 micromolar, 125 micromolar to 300 micromolar, 150 micromolar to 300 micromolar, 175 micromolar to 300 micromolar, 200 micromolar to 300 micromolar, 225 micromolar to 300 micromolar, 250 micromolar to 300 micromolar or 275 micromolar to 300 micromolar. When concentrations of sandalwood oil are expressed in micromolarity, it is understood that this is the micromolarity of alpha-santalol in the sandalwood oil. For example, the concentration of sandalwood oil can be about 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 60, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 295, 300 micromolar or any concentration in between the concentrations set forth herein. **The concentration of sandalwood in the composition can also be about 0.0002% to about 0.1% sandalwood oil (w/w).** For example, the concentration of sandalwood in the composition can be about 0.0002% to about 0.005%, or from about 0.0002% to about 0.01%, or from about 0.0002% to about 0.05%, or from about 0.0002% to about 0.1%. The concentration of sandalwood oil can also be about 0.0002%, 0.0003%, 0.0004%, 0.0005%, 0.0006%, 0.0007%, 0.0008%, 0.0009%, 001%, 0.002%, 0.003%, 0.004%, 0.005%, 0.006%, 0.007%, 0.008%, 0.009%, 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.1% or any percentage in between the percentages set forth herein. **Multiple administrations and/or dosages can also be used. Effective doses can be extrapolated from dose-response curves derived from in vitro or animal model test systems.**

(emphasis added).

Accordingly, the claim language broadly recites “an effective amount of a composition.” The claim language does not place a limit on the effective amount of the composition. It only limits the concentration (in % (w/w)) of sandalwood oil in this composition, but not the effective amount of the composition. In this line, the specification similarly discloses that this effective amount can be any amount, depending on factors, such as the nature of the cancer and its associated symptoms, the type of cancer, the route of administration, and the seriousness of the disease or disorder. The specification further recites that this effective amount can be determined by standard clinical techniques, that *in vitro* assays can be employed to identify optimal dosage ranges, that the precise dose to be employed in the formulation will also depend on, and should be decided according to the judgment of the practitioner and each subject's circumstances, that multiple administrations and/or dosages can also be used, and that effective doses can be extrapolated from dose-response curves derived from in vitro or animal model test systems. Thus, the specification again confirms that the claim does not place a limit on the effective amount of the composition. It only limits the concentration (in % (w/w)) of sandalwood oil in this composition, but not the effective amount of the composition.

In view of this clear and explicit language of Appellant's claims, all of Appellant's arguments over the prior art are misconstructions at best. For instance, Appellant has argued as to Haque: "Haque is unclear as to the concentration of the α- and β-santalols used in treating warts. Haque merely notes that santalols constitute over 65% of sandalwood oil and describes the use of a mixture of isolated α- and β-santalols. [...] The Examiner accordingly erred in her characterization of Haque as disclosing treatment of skin and cervical cancer with an effective amount of sandalwood oil." In response, to the contrary, Haque discloses use of sandalwood oil, and/or its isolated active constituents, in a composition for treating cancer, i.e. in effective amounts for treating cancer. Further, any and all of the references cited, in addition to Haque, disclose use of sandalwood oil, and/or its isolated active constituents, for treating cancer, i.e. in effective amounts for treating cancer.

Appellant's arguments over Banerjee fail for a similar reason. Appellant has argued, *inter alia*, that: "Banerjee discloses that sandalwood oil from S. album, given by oral administration to mice, results in increased glutathione S-transferase (GST) activity. These results were obtained by orally feeding 5 μl or 15 μl of 100% sandalwood oil to the mice. By using 5 μl or 15 μl of 100% sandalwood oil for 10 or 20 days, Banerjee determined the volume of 100% sandalwood oil and the duration of treatment that could be used to enhance GST activity and/or sulfhydryl group (SH) levels. This is not equivalent to optimizing a range of concentrations of a composition for use for treating non-skin cancer. Nothing in Banerjee discloses or suggests that increased glutathione S-transferase (GST) activity or SH levels would be observed at a concentration other than 100% sandalwood oil, much less at a concentration of 0.0002-0.1% sandalwood oil (w/w)." Again, what Appellant has argued is contrary to Appellant's own claim language and support in the specification. Per Appellant's own specification, an effective amount can be any amount, depending on factors, such as the nature of the cancer and its associated symptoms, the type of cancer, the route of administration, and the seriousness of the disease or disorder. For instance, 5 μl of 100% sandalwood oil (according to Banerjee) would theoretically contain the same effective amount as, for instance, 50 μl of 0.1% sandalwood oil (according to Appellant's claims).

Moreover, contrary to Appellant's arguments, Banerjee discloses motivation to optimize the therapeutic amount. It discloses with respect to doses that the study demonstrated that oral gavage treatment led to significantly higher SH content in animals treated with both dose levels for 10 days, whereas the increase seen with treatment at 15 µl (versus 5 µl) was significant in only the 10-day treatment group. (p. 108, col. 1). Thus, in Banerjee the same compound is used for the same indication, only in a more concentrated form than in Appellant's claims. This clearly creates a *prima facie* case of obviousness, as it demonstrates that there is no criticality in the concentration amount, *per se*, as both a 100% concentrated amount of sandalwood oil similarly shows treatment of the same compound for the same disease/ condition. Appellant has nowhere rebutted this with a showing of unexpected results of a concentration amount of 0.0002-0.1% sandalwood oil (w/w), particularly where as here Appellant's claim is directed to any effective amount of the composition.

Appellant has further no support for the following argument: "The Examiner further suggests that Banerjee stands for the proposition that 'not all doses studied were found to be efficacious for the duration of treatment.' See Final Office Action dated April 22, 2014, page 9. This assertion is factually incorrect." In response, the Examiner stands behind the exact quote from the final office action: "With respect to doses, the study demonstrated that this oral gavage treatment led to significantly higher SH content in animals treated with both dose levels for 10 days, whereas the increase seen with treatment at 15 µl was significant in only the 10-day treatment group. (p. 108, col. 1)." The following is an excerpt in support from Banerjee:

> The hepatic acid soluble SH content was found to be significantly higher in animals treated with 5 μl of sandalwood oil at both the dose levels for 10 days ($P < 0.001$) whereas the increase in SH levels seen at 15 μl dose level was significant in only the 10-day-treatment groups.

The Examiner further stands behind her statements in the final office action on pages 5 and 7, which provide: "Thus, Banerjee alone discloses an additional specific cancer, per Applicant's claims 6-8." (p.5); "A person of skill in the art would have been motivated to do so as Haque shows sandalwood oil to treat different types of cancer- e.g. of the skin and of the cervix on topical application, and Banerjee further suggests that oral administration of sandalwood oil has a chemopreventative action on carcinogenesis, as exemplified from results obtained from liver tissue (i.e. non-skin related)." (p. 7). Indeed, the entirety of the disclosure of Banerjee relates to oral administration of sandalwood oil and its chemopreventative action on carcinogenesis, as exemplified from results obtained from liver tissue (i.e. non-skin related). It is credibility defeating for Appellant to be stating that: "[t]he only specific cancer mentioned in Banerjee, however, is DMBA-induced skin papillomagenesis, a chemically induced skin cancer. *See* Banerjee, page 109." Indeed, Banerjee makes it explicit that the referenced by Appellant section is in fact one, which states that in view of the studies in the liver it is possible that sandalwood can act as a chemopreventative agent against chemical carcinogenesis, that it is in reference to unpublished data (i.e. <u>not</u> the Banerjee reference), and that further work is being done "on this aspect <u>using a different</u>

model system." (emphasis added). The following is an excerpt in support from Banerjee:

> In view of the fact that sandalwood oil can induce the activity of GST and the level of SH group in liver it is possible that it can act as a chemopreventive agent against chemical carcinogenesis. Indeed our experiments (unpublished data) have suggested this property in DMBA-induced skin papillomagenesis. Further work on this aspect using a different model system is in progress in our laboratory.

Appellant's arguments against Piggott alone as not teaching the instant method are further not found to be persuasive. In response against Appellant's argument against the references individually, it is noted that one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references. *See In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981); *In re Merck & Co.*, 800 F.2d 1091, 231 USPQ 375 (Fed. Cir. 1986). In the instant case, Piggott was only used, as explicitly stated in the office action, for its teachings of distilled or extracted sandalwood oil.

Appellant's arguments against Azam, Va gasena and Thiruvalluvar gnana vettian are further not found to be persuasive. First, against Appellant's arguments that these references disclose "complex mixtures of a wide variety of components", it is noted that Appellant's transitional phrase "comprising" is open-ended, and as such does not preclude the presence of other ingredients. Second, the references explicitly disclose treatment of cancer. Third, statements by Appellant that one of skill in the art "might view" these compositions as "sacred unguents", or "may expect" that sandalwood oil was used as a perfume or flavorant, are arguments by counsel that are speculative at best, and do not find support in the explicit language of the references. If a *prima facie* case of obviousness is established, the burden shifts to the applicant to come forward with arguments and/or evidence to rebut the *prima facie* case. *See, e.g., In re Dillon,* 919 F.2d 688, 692, 16 USPQ2d 1897, 1901 (Fed. Cir. 1990). However, arguments of counsel cannot take the place of factually supported objective evidence. See, e.g., *In re Huang,* 100 F.3d 135, 139-40, 40 USPQ2d 1685, 1689 (Fed. Cir. 1996); *In re De Blauwe,* 736 F.2d 699, 705, 222 USPQ 191, 196 (Fed. Cir. 1984). *See also* MPEP 2145.

> b) Appellant argues that "[t]he cited references fail to provide an apparent reason to combine the references to arrive at the claimed method" (Ap. Br. at 8-10).

Appellant's arguments have been considered, but not found to be persuasive. The Examiner repeats, and incorporates by references her responses from above.

First, it is noted that the office action properly rationale to combined. Specifically, it provides in relevant part as follows. "Although both Haque and Banerjee do not explicitly disclose the particular amounts claimed by Appellants, per Appellant's claims as amended, it would have been further obvious to a person of skill in the art to optimize these amounts, in order to arrive at Appellant's claimed range in % (w/w). A person of skill in the art would have been motivated to do so in order to enhance the therapeutic efficacy. Generally, mere optimization of ranges will not support the patentability of subject matter encompassed by the prior art unless there is evidence indicating such concentration or temperature is critical. "When the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimal or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955); "The normal desire of scientists or artisans to improve upon what is already generally known provides the motivation to determine where in a disclosed set of percentage ranges is the optimum combination of percentages." *In re Peterson*, 315 F.3d 1325, 1330 (Fed. Cir. 2003). It has been held that it is within the skills in the art to select optimal parameters, such as amounts of ingredients, in a composition in order to achieve a beneficial effect. *In re Boesch*, 205 USPQ 215 (CCPA 1980). MPEP 2114.04. Moreover, motivation to do so is additionally found in Banerjee, which shows a study with optimizing such ranges, as expressed in μl. A person of skill in the art would have been motivated to estimate such ranges in other units as well (such as claimed by Appellant), guided by the disclosure in Banerjee that not all doses studied were found to be efficacious for the duration of treatment, with the lower dose showing better efficacy. A person of skill in the art would have been further motivated to do so guided by the disclosure in Piggott, which shows that it is essential for distilled or extracted sandalwood oil to determine such therapeutic ranges, as depending on the technique of extraction used, and the portion of wood used, the amounts of the active ingredients in sandalwood vary significantly. Thus, the skilled artisan would have had at its disclosure at the time of the invention multiple sources independently of one another, and complimentary to one another, establishing motivation to optimize the dose range of sandalwood oil in practicing the claimed method."

Moreover, to this day Applicant has failed to make any showing whatsoever of criticality of the claimed ranges. "The law is replete with cases in which the difference between the claimed invention and the prior art is some range or other variable within the claims. . . . In such a situation, the applicant must show that the particular range is critical, generally by showing that the claimed range achieves unexpected results relative to the prior art range." *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). *See* MPEP § 716.02 - § 716.02(g) for a discussion of criticality and unexpected results.

Further, even if one were not to consider, *arguendo*, an optimization rationale *per se*, it is noted based on Haque it would have been *prima facie* obvious to apply α- and β-santalols- the active ingredients in sandalwood oil- in a broad range of "a small amount", including amounts in the claimed range of “about 0.0002-0.1% sandalwood oil (w/w).” Per Haque: “A further embodiment of the method of this invention comprises washing the affected area of the body with the soap, rinsing the area, placing ***a small amount*** of the α- and β-santalols on the tumor to be treated and then gently rubbing to facilitate penetration… The present invention is based, in part, on the discovery that a commercially available soap manufactured by the aforementioned companies may be useful in the treatment of viral-induced tumors in humans. Specifically, the invention is directed to the discovery that sandalwood oil is the active component of the soap and more specifically, that α- and β-santalols are the active ingredients in sandalwood oil.” (col. 5, ll. 59-63; emphasis added). Accordingly, it would have been *prima facie* obvious to apply α- and β-santalols in a broad range of "a small amount", including amounts in the claimed range of "about 0.0002-0.1% sandalwood oil (w/w)."

Thus, based on the above and contrary to Appellant's arguments, the Examiner properly relied on the holding of *In re Peterson*, and Appellant has not presented any persuasive argument that none of the cited references disclose or suggest a set of percentage ranges from which an optimal range or a lower limit could have been determined. Moreover, Appellant's reliance on *Ex parte Whalen*, 89 USPQ2d 1078 (BPAI 2008) is wholly inappropriate, and distinguishable on the instant facts where the issue is not at all optimizing viscosity and molecular weight of biocompatible polymer wherein low viscosity is a desirable characteristic, but is rather applying a small concentration in any effective amount of a composition. To the contrary of Appellant's arguments, the art here, to take Haque alone as a single non-limiting example, discloses applying a small amount of the α- and β-santalols on the tumor to be treated. Regarding Banerjee, the Examiner already addressed above that the reference to "chemically induced skin cancer" refers to another study, not the one disclosed in Banerjee, and that with respect to doses, Banerjee demonstrates that oral gavage treatment leads to significantly higher SH content in animals treated with both dose levels (5 μl or 15 μl) for 10 days, whereas the increase seen with treatment at 15 μl was significant in only the 10-day treatment group. (p. 108, col. 1). Further, Appellant has not adequately explained why the Examiner's positions regarding Azam, Va gasena and Thiruvalluvar gnana vettian are incorrect.

c) Appellant argues that "[t]he cited references fail to provide one of ordinary skill in the art with a reasonable expectation of success" (Ap. Br. at 10-11).

Appellant's arguments have been considered, but not found to be persuasive. The Examiner repeats, and incorporates by references her responses from above.

Appellant's reliance on *Alcon Research, Ltd v. Apotex Inc.*, 687 F.3e 1362 (Fed. Cir. 2012) is clearly inappropriate on the instant facts. In *Alcon Research*, as Appellant itself noted "since the prior art disclosed only concentrations up to 0.01%, a person of ordinary skill in the art would not have a reasonable expectation of success for increasing the dosage to 0.15 w/v". (Ap. Br. at 11). To the contrary of Appellant's arguments, the art here, to take Haque alone as a single non-limiting example, discloses applying a small amount of the α- and β-santalols on the tumor to be treated. Accordingly, it would have been *prima facie* obvious to apply α- and β-santalols in a broad range of "a small amount", including amounts in the claimed range of "about 0.0002-0.1% sandalwood oil (w/w)."

B) This application is further not patentable, as to this date Appellant has never provided any evidence of unexpected results, or <u>of other secondary indicia of non-obviousness</u>

This application is further not patentable, as to this date Appellant has never provided any evidence of unexpected results, or of other secondary indicia of non-obviousness. More need not be said.

In sum, this is a case with a very strong factual and legal record against Appellant's claims. Accordingly, the Examiner respectfully requests that the Patent Trial and Appeal Board uphold the rejections.

/SVETLANA M. IVANOVA/

Primary Examiner, Art Unit 1627

Conferees:

/SREENI PADMANABHAN/

Supervisory Patent Examiner, Art Unit 1627

/Jon D. Epperson/

Primary Examiner

Requirement to pay appeal forwarding fee. In order to avoid dismissal of the instant appeal in any application or ex parte reexamination proceeding, 37 CFR 41.45 requires payment of an appeal forwarding fee within the time permitted by 37 CFR 41.45(a), unless appellant had timely paid the fee for filing a brief required by 37 CFR 41.20(b) in effect on March 18, 2013.